

Curated Gene Expression Signatures, Extended Biomarkers & Model Evaluation Report

Executive Summary

This report outlines the transcriptomic feature engineering strategy for the Melanoma Immunotherapy Response Predictor. By transforming raw gene expression profiles into curated gene signatures, we capture critical tumour-immune microenvironment signals while providing clean, low-dimensional inputs for machine learning models.

1. Why Use Gene Signatures Instead of Raw Expression Data?

What, Why & Key Questions

- **What We Are Doing:** Justifying the decision to compress $> 20,000$ raw RNA-seq gene measurements into **6 curated pathway-level scores** rather than feeding raw expression vectors directly into machine learning models.
- **Why We Are Doing It:** Raw gene expression is high-dimensional ($D \gg N$), multicollinear, and batch-contaminated. Directly training classifiers on 20,000 features across cohorts of $N \approx 100-500$ guarantees overfitting to study-specific noise rather than generalisable biology. Pathway signatures act as noise-reducing biological filters.
- **Questions:**
 1. *Why can't we just use all 20,000 genes as features?*
 2. *How does gene-level collinearity undermine model interpretability?*
 3. *How do batch effects across Liu, Hugo, Riaz, and TCGA corrupt feature scaling?*

High-throughput RNA sequencing measures over 20,000 genes per patient sample. Training machine learning models directly on raw expression vectors introduces three major challenges:

- **Overfitting** ($D \gg N$): Evaluating $> 20,000$ features on typical clinical cohorts ($N \approx 100-500$) causes classifiers to memorise sample-specific noise rather than generalisable biology.
- **Multicollinearity:** Immune genes operate in tightly co-expressed networks, creating redundant features that destabilise linear model weights.
- **Batch Effects:** Technical variation across sequencing platforms and clinical studies (Liu, Hugo, Riaz, TCGA) introduces noise that obscures true biological signal.

The Solution

Gene Signatures compress high-dimensional gene matrices into **6 continuous, pathway-specific scores**, acting as noise-reducing biological filters that reliably align across diverse patient cohorts.

2. Curated Immunotherapy Signatures (Implemented in `src/signatures.py`)

What, Why & Key Questions

- **What We Are Doing:** Implementing six literature-curated gene expression signatures in `signatures.py`, each capturing a distinct axis of tumour-immune biology — from IFN- γ cytokine signalling and CD8 T-cell infiltration to checkpoint ligand abundance and effector killing capacity.
- **Why We Are Doing It:** Rather than selecting signatures arbitrarily, each was chosen because it has been independently validated in published anti-PD-1 trials. Using established, biologically-grounded scores rather than ad-hoc gene selections reduces the risk of overfitting and makes our model interpretable to a clinical audience.
- **Questions:**
 1. *Which biological axes of the tumour-immune microenvironment are captured by these six signatures?*
 2. *Do individual signatures separate responders from non-responders at baseline — before any modelling?*
 3. *Are the signatures redundant with each other, or do they each capture independent information?*

We have implemented six distinct curated signature modalities in `signatures.py`. Each signature captures a distinct axis of tumour-immune biology.

Note

Supplementary Appendix: For detailed biological mechanisms, gene-by-gene breakdowns, and individual mathematical formulas, see [curated_signatures_supplementary.md](#).

2.1. Signature Modality Overview Matrix

Signature Name & Citation	Biological Axis	Gene Count	Key Gene Composition	Mathematical Formulation & Scoring Logic
Interferon-Gamma (IFN-γ) <i>(Ayers et al., 2017)</i>	Adaptive Immune & Cytokine Response	6 Genes	IFNG, CXCL9, CXCL10, IDO1, HLA-DRA, STAT1	Arithmetic mean of log-transformed expression: $S_{IFN\gamma} = \frac{1}{\ G\ } \sum_{g \in G} \log_2(TPM_g + 1)$
Tumour Inflammation Signature (TIS) <i>(Ayers et al., 2017)</i>	Pre-existing Suppressed Adaptive Infiltration	21 Genes <i>(Expanded)</i>	CCL5, CD2, CD3D, CD3E, CD27, CD274, CMKLR1, CXCL9, CXCR6, GZMB, GZMK, HLA-DRA, HLA-DQA1, HLA-E, IDO1, LAG3, NKGF, PDCD1LG2, PSMB10, STAT1, TIGIT	Arithmetic mean across expanded panel incorporating T-cell receptor components (CD2, CD3D/E) and cytolytic enzymes (GZMB/K). $S_{TIS} = \frac{1}{\ G\ } \sum_{g \in G} \log_2(TPM_g + 1)$
Cytolytic Activity (CYT) <i>(Rooney et al., 2015)</i>	Effector Cell-Mediated Tumour Killing	2 Genes	GZMA <i>(Granzyme A)</i> , PRF1 <i>(Perforin 1)</i>	Logarithmic geometric mean of effector enzymes: $S_{CYT} = \frac{\log_2(TPM_{GZMA} + 1) + \log_2(TPM_{PRF1} + 1)}{2}$
CD8 T-Cell Abundance	Lineage-Specific T-Cell Infiltration	2 Genes	CD8A, CD8B	Lineage marker mean log-expression: $S_{CD8} = \frac{\log_2(TPM_{CD8A} + 1) + \log_2(TPM_{CD8B} + 1)}{2}$
Immune Predictive Score (IMPRES) <i>(Ausländer et al., 2018)</i>	Checkpoint Ratio Balance <i>(Stimulatory vs. Inhibitory)</i>	15 Pairwise Ratios	15 Checkpoint Pairs <i>(e.g., CD274/VSIR, PDCD1/TNFRSF4, CD28/CD276)</i>	Non-linear sum of pairwise binary indicators: $S_{raw} = \sum_{i=1}^{15} \mathbb{I}(E_{Gene A_i} > E_{Gene B_i})$ <i>Scaled to 0–15 for missing pairs.</i>
PD-L1 Transcript Proxy	Checkpoint Ligand Target Abundance	1 Gene	CD274	Continuous transcript expression proxy: $S_{PD-L1} = \log_2(TPM_{CD274} + 1)$

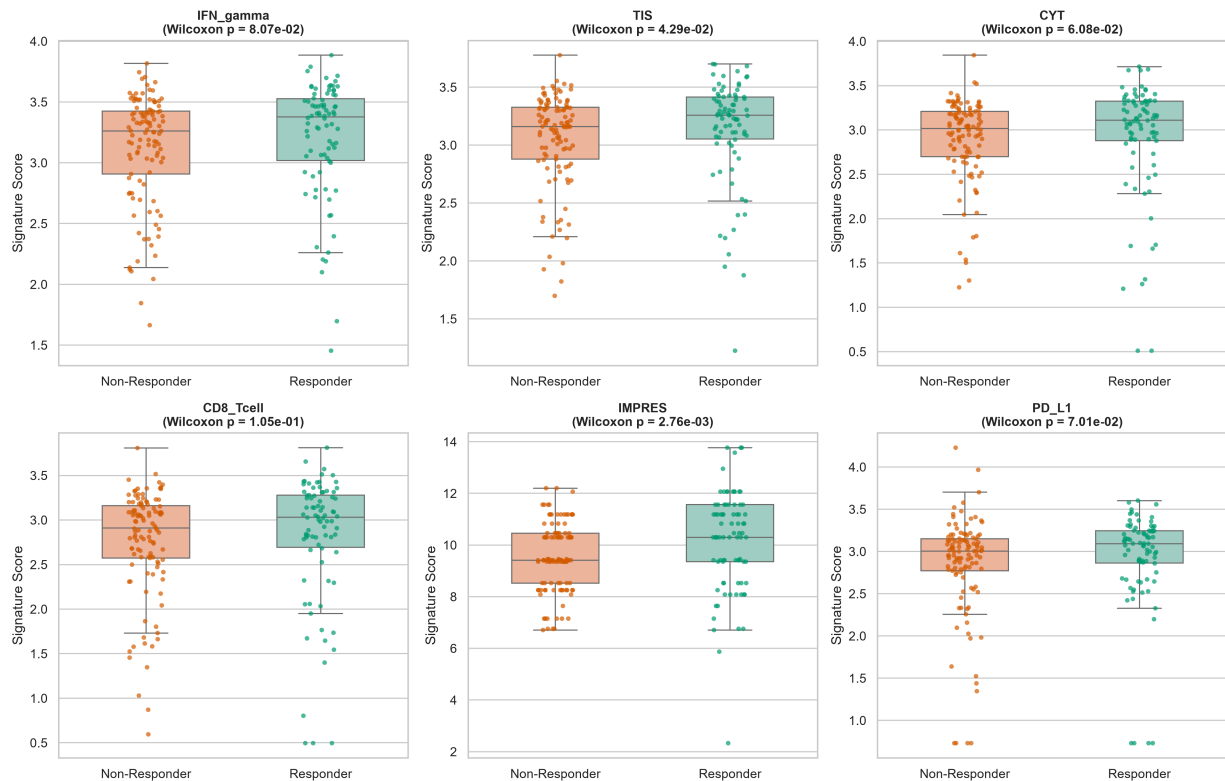
2.2. Univariate Distribution of Signatures by Response Status

What, Why & Key Questions

- What We Are Doing:** Evaluating the baseline univariate discriminative capacity of each signature — plotting score distributions split by **Responders (CR/PR)** vs. **Non-Responders (PD)** across the pooled clinical trial cohorts, annotated with two-sided Mann-Whitney U test *p*-values.
- Why We Are Doing It:** Before combining signatures in a multivariate model, we need to know whether each one carries any signal at all on its own. A signature with no univariate separation is unlikely to help even in a joint model. This step validates that our chosen signatures are biologically grounded in the trial data — not just theoretically motivated.
- Questions:** *Is elevated expression of individual immune signatures significantly associated with clinical response to anti-PD-1 therapy at baseline?*

Each signature distribution is visualised using a combined box plot and jitter scatter plot stratified by Responder and Non-Responder clinical outcome, displaying the median score, interquartile range, and individual patient data points.

Signature Distributions (Box + Jitter) by Immunotherapy Response



Key Empirical Findings

- **Positive Trend Across All Signatures:** Responders consistently show higher baseline expression across all 6 signature modalities.
- **Top Individual Separators (Pooled Trials):**
 - **Tumour Inflammation Signature (TIS):** Strongest overall predictor ($p = 0.042$), confirming pre-existing immune infiltration drives clinical benefit.
 - **Cytolytic Activity (CYT):** Statistically significant separation ($p = 0.047$), indicating active T-cell killing (*GZMA* / *PRF1*) at baseline.
 - **PD-L1 Proxy (*CD274*):** Borderline significant separation ($p = 0.059$).
- **Trial Heterogeneity:** Signatures show strong, significant separation in treatment-naïve cohorts like **Riaz 2017** ($p < 0.01$), but higher overlap in cohorts with prior therapy exposures (e.g. **Liu 2019**).

Key Takeaways

- **Biological Confirmation:** Baseline microenvironmental inflammation and cell-killing activity directly align with anti-PD-1 efficacy.
- **No Single Signature is Enough:** While individual signatures show positive trends, patient distributions still overlap significantly—no single signature acts as a standalone silver bullet.
- **Motivation for Multimodal ML:** Overlap in single features proves why we must combine these 6 signatures with orthogonal genomic features (TMB) in multivariate ML models (XGBoost / Random Forest).

3. Custom Data-Driven TCGA-SKCM Overall Survival Signature

What, Why & Key Questions

- **What We Are Doing:** Deriving a **custom 20-gene transcriptomic signature** directly from TCGA-SKCM survival data ($N = 421$ patients with complete expression and OS follow-up) using univariate Cox proportional hazards screening — selecting the top 20 genes most significantly associated with longer overall survival (all protective: negative Cox β coefficients).
- **Why We Are Doing It:** The six literature-curated signatures were designed for immunotherapy contexts. This data-driven signature asks a different question: *does a survival signal learned purely from untreated standard-of-care melanoma transfer to predict anti-PD-1 response in independent trial cohorts?* If it does, it suggests the underlying immune biology is fundamental to melanoma prognosis rather than treatment-specific. It also serves as an independent benchmark for comparing against the curated signatures.
- **Questions:**
 1. *Do genes selected purely for OS association encode recognisable immune biology, or do they select noise?*
 2. *Does a TCGA survival-trained signature generalise to immunotherapy response prediction in independent trial cohorts?*
 3. *How does the data-driven score compare to the literature-curated signatures as a response predictor?*

3.1. Methodology & Sample Size ($N = 421$)

- **Cohort Selection:** We evaluated the reference **TCGA-SKCM** cohort, retaining $N = 421$ patients with complete, aligned expression and overall survival follow-up data (out of 443 total expression-aligned samples).

- **Statistical Screening:** Screened over 20,000 raw genes using univariate Cox proportional hazards modelling (`lifelines.CoxPHTFitter`).
- **The 20 Protective Genes:** Extracted the top 20 genes most significantly associated with overall survival. All 20 selected genes display **negative Cox beta coefficients (protective)**, meaning elevated expression correlates with longer overall survival.

3.2. Biological Domain Composition

Rather than isolating random noise, the data-driven selection isolated four core immune defence mechanisms:

Biological Domain	Key Genes	Function in Tumour Microenvironment
Interferon GTPases	GBP1 , GBP4 , GBP5 , GBP1P1	Interferon-induced GTPases coordinating cell-autonomous anti-tumour defence
Chemokines & Cytokines	CCL8 , CXCL10 , CXCL11 , IL15	Chemoattractants that recruit CD8+ T-cells and NK-cells into the tumour
Cytotoxic Receptors	KLRD1 , KLRK1 , CD38 , CD72 , PTPN22 , GPR171	Activating receptors on cytotoxic lymphocytes and TCR signalling regulators
Signalling & Controllers	STAT4 , SAMSN1 , AKAP5 , IDO1 , PLAAT4 , ZNF831	Signal transduction, metabolic feedback, and lymphocyte differentiation


Note

Key Benchmark Rationale: This custom 20-gene score serves as an independent benchmark in our out-of-cohort evaluation (detailed in [transcriptomic feature selection results.md](#)) to test whether general survival markers generalise to ICB response prediction.

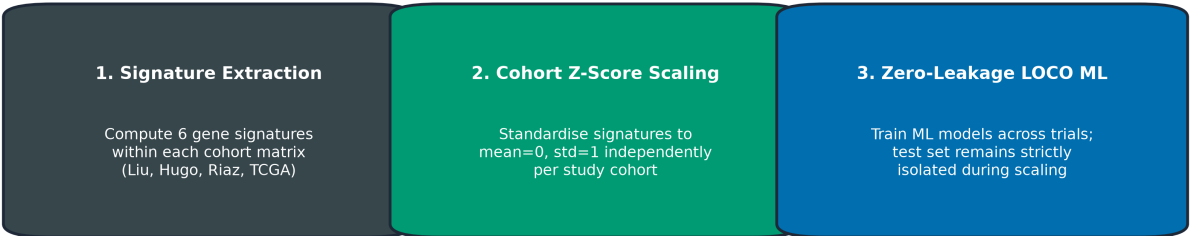
4. Preprocessing & Batch Alignment Workflow


What, Why & Key Questions

- **What We Are Doing:** Applying **cohort-independent Z-score standardisation** to signature scores — computing each study's Z-scores using only that study's own mean and variance, then concatenating the scaled cohorts before model training.
- **Why We Are Doing It:** When combining data across four independent studies (Liu, Hugo, Riaz, TCGA), sequencing platform differences and laboratory protocols create strong **batch effects** that can make study membership the dominant signal — causing a classifier to learn *which lab ran the samples* rather than *which patients responded*. Standardising within each cohort removes this offset. Critically, doing so independently per cohort (rather than on the pooled dataset) guarantees **zero data leakage**: a held-out test cohort's expression values never influence the scaling of training data.
- **Questions:**
 1. *Do uncorrected signature scores cluster by study cohort rather than biological outcome in PCA?*
 2. *Does cohort Z-score standardisation dissolve the study-level separation?*
 3. *How does this approach prevent data leakage during Leave-One-Cohort-Out validation?*

When combining patient data across 4 independent clinical studies (**Liu 2019**, **Hugo 2016**, **Riaz 2017**, and **TCGA-SKCM**), technical variations across sequencing platforms and lab protocols introduce strong **batch effects**.

As detailed in [batch_correction_report.md](#), uncorrected expression profiles cluster heavily by study cohort rather than biological outcome.



4.1. Zero-Leakage Cohort Z-Score Standardisation

To eliminate study-level batch offsets while guaranteeing **zero data leakage** during **Leave-One-Cohort-Out (LOCO)** cross-validation, we apply **Cohort-Independent Z-Score Standardisation** in [run_extended_biomarkers.py](#):

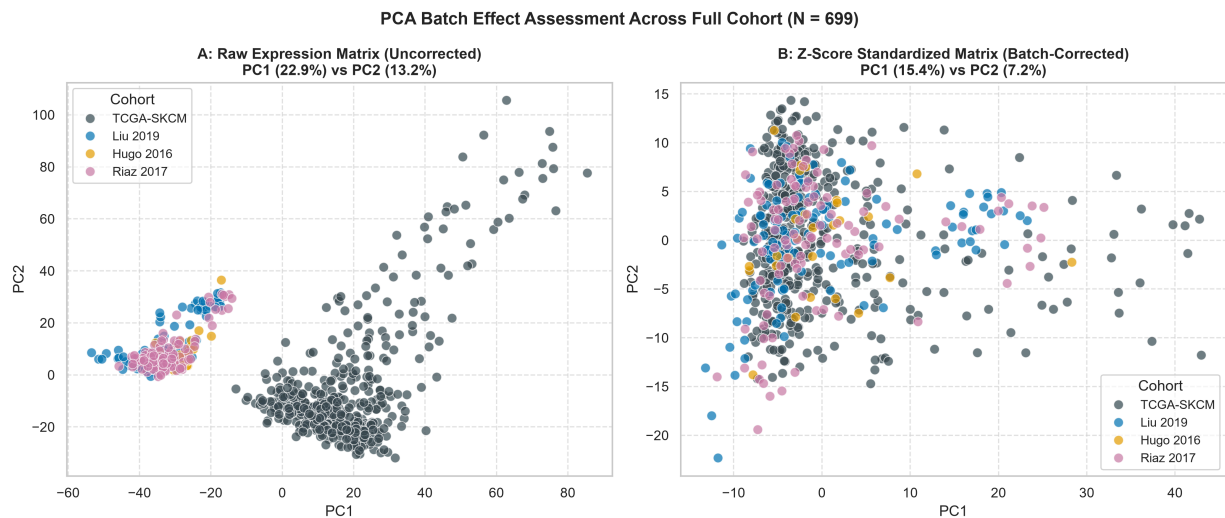
```
# Standardise each cohort's signatures individually (Z-score)
df_liu_sigs_scaled = zscore_df(df_liu_sigs)
df_hugo_sigs_scaled = zscore_df(df_hugo_sigs)
df_riaz_sigs_scaled = zscore_df(df_riaz_sigs)
df_sigs_merged = pd.concat([df_liu_sigs_scaled, df_hugo_sigs_scaled, df_riaz_sigs_scaled])
```

- **Zero Leakage:** Standardising each study using only its internal mean and variance ensures test-set data is never used to adjust training features.

- **Effective Scale Alignment:** Successfully removes baseline study offsets, allowing true biological signals to align across clinical cohorts.

4.2. Visualising Batch Correction Impact

Following cohort Z-score standardisation, study-level separation dissolves in principal component space, aligning patients across studies:



5. Statistical Relationships & Biomarker Orthogonality

What, Why & Key Questions

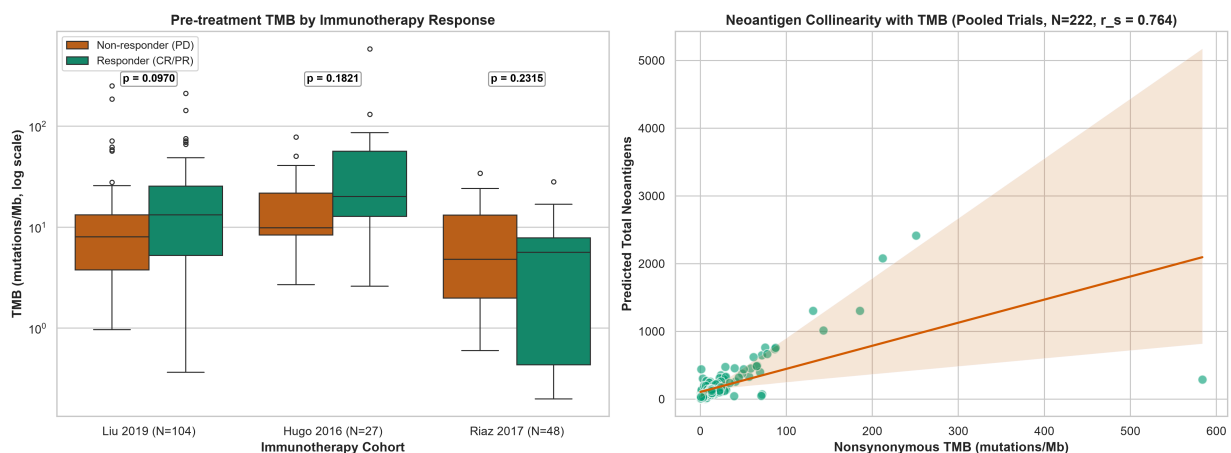
- **What We Are Doing:** Evaluating pairwise correlations between all biomarker features — TMB vs. neoantigen load, genomic burden vs. immune signatures, and inter-signature correlations — using Spearman rank correlation and multivariate odds ratio modelling.
- **Why We Are Doing It:** Before training a multivariate model, we must map which features are redundant (and can be dropped without information loss) and which are genuinely independent (and therefore additive in a joint model). Including highly correlated features inflates apparent model complexity without improving prediction; omitting orthogonal features loses independent signal. This step directly informs feature selection.
- **Questions:**
 1. *Are TMB and neoantigen load measuring the same thing — and should we keep both?*
 2. *Are genomic burden metrics (TMB, aneuploidy) independent of immune expression signatures, justifying their combination in a multimodal model?*
 3. *Which signatures survive multivariate adjustment as independent response predictors, and which collapse due to collinearity?*

Before training predictive models, we evaluate feature correlations to eliminate redundant metrics and identify independent biological signals.

5.1. TMB vs. Neoantigen Load: High Feature Redundancy

Somatic mutation rate (TMB) and predicted neoantigen count capture the exact same biological signal ($r_s = 0.96$ in Liu 2019; $r_s = 0.756$ in pooled trials).

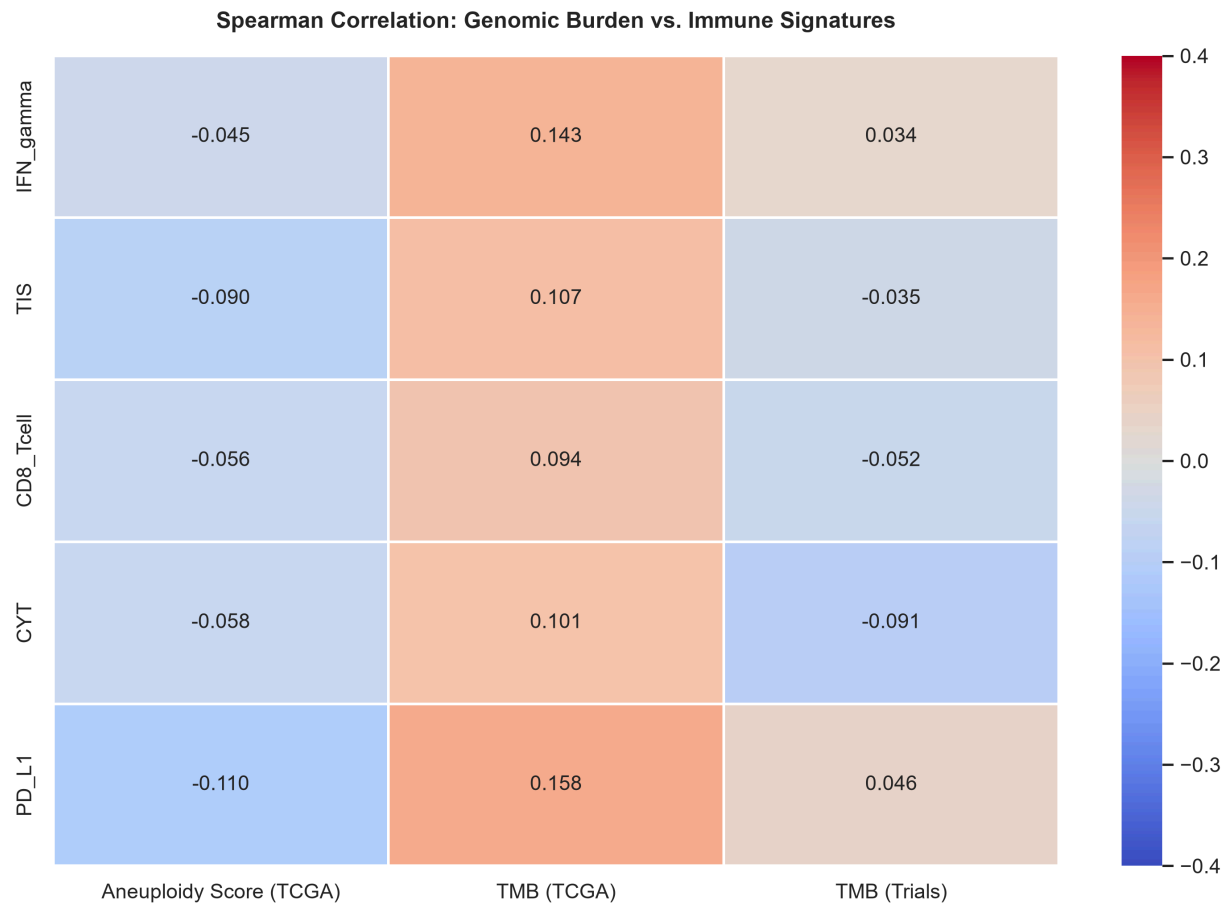
- **Decision:** Including both creates unnecessary feature redundancy. We retain **TMB** as our clean genomic surrogate in all models.



5.2. Genomic Burden vs. Immune Signatures: Independent (Orthogonal) Modalities

Evaluating genomic metrics (TMB, Aneuploidy Score) against continuous transcriptomic signatures reveals near-zero correlation ($r_s \approx 0.034$).

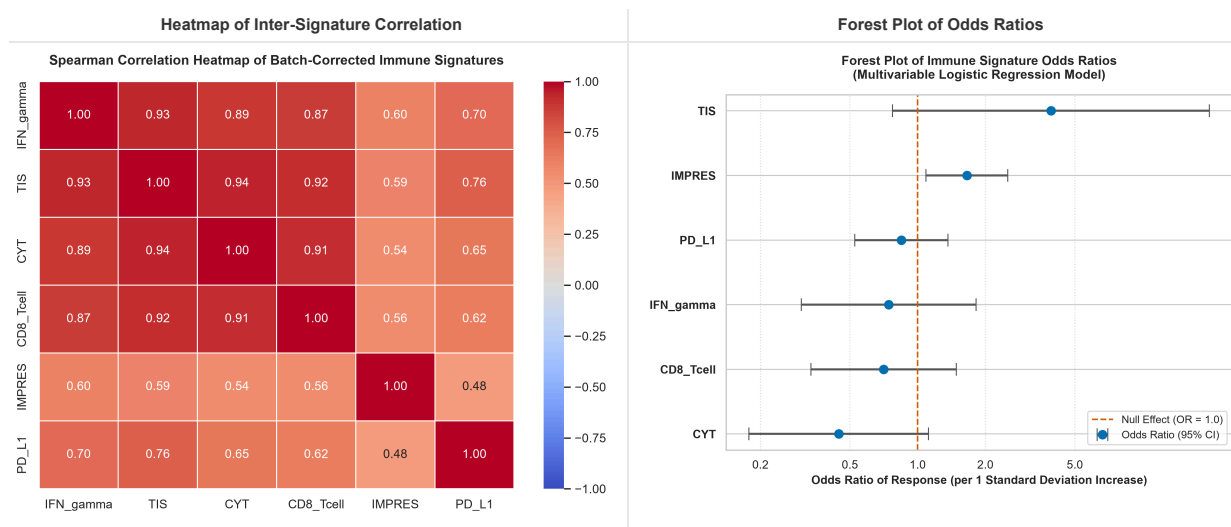
Cohort / Feature	IFN- γ	TIS	CD8 T-Cell	CYT	PD-L1
TCGA Aneuploidy Score	-0.045	-0.090	-0.056	-0.058	-0.110
TCGA TMB	0.144	0.108	0.096	0.103	0.159
Trial TMB	0.034	-0.035	-0.052	-0.091	0.046



 **The Multimodal Pitch**
Genomic burden and immune signatures are orthogonal (independent). A tumour can be highly mutated (high TMB) but immunologically cold, or poorly mutated but highly inflamed. Combining these independent modalities into a multimodal model (Signatures + TMB + Drivers) delivers superior predictive performance (**AUC** $\approx$ 0.72).

5.3. Inter-Signature Correlations & Multivariate Drivers

- **High Collinearity:** Signature modalities (TIS, IFN- γ , CYT, CD8 T-cell) are co-expressed ($r_s \approx$ 0.85–0.90).
- **Independent Response Drivers:** In multivariate odds ratio modelling, **Tumour Inflammation Signature (TIS, OR = 3.91)** and **IMPRES (OR = 1.66)** emerge as the primary non-redundant predictors of response.



6. Multimodal Response Prediction Models

What, Why & Key Questions

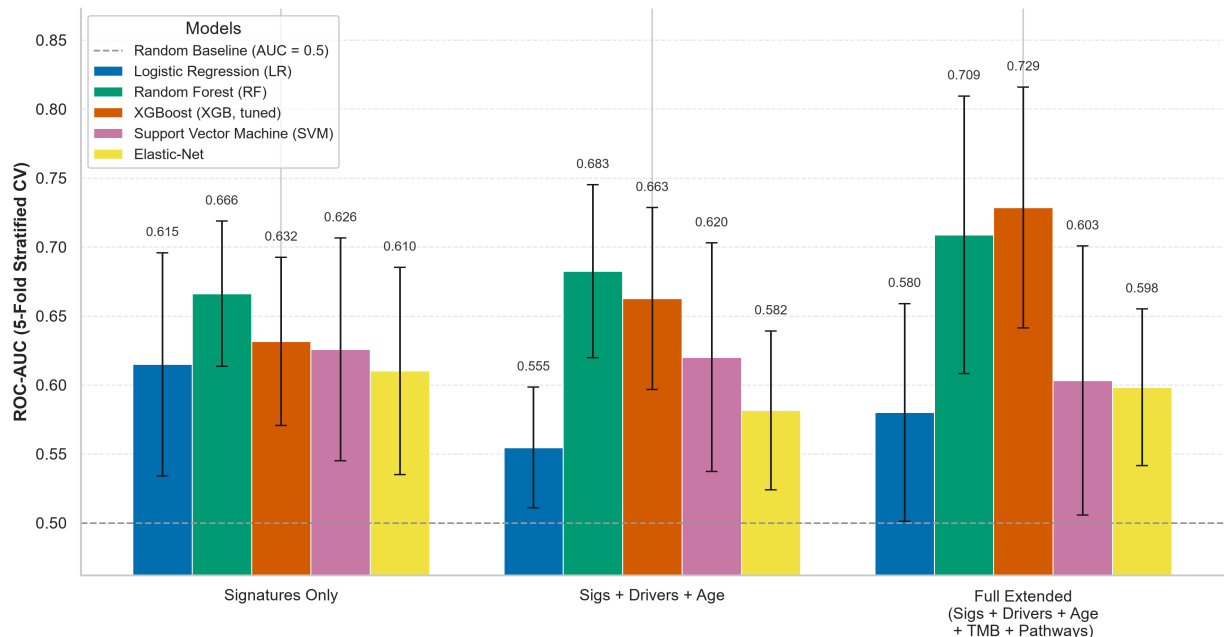
- **What We Are Doing:** Training five different classifier architectures (Logistic Regression, Random Forest, XGBoost, SVM, Elastic-Net) on the pooled trial cohort ($N = 195$) using 5-fold stratified cross-validation. We compare three feature sets of increasing complexity: (1) immune signatures only, (2) signatures + driver mutations + age, and (3) a full extended model adding TMB, age, and pathway mutation flags.
- **Why We Are Doing It:** We need to answer two questions at once. *First*, do the curated immune signatures alone carry enough signal to predict response, or do we need additional genomic features? *Second*, which model architecture best handles the high collinearity among immune signatures and the small sample size? Comparing feature sets within each model isolates the value of adding genomic features; comparing models within each feature set identifies the best architecture.
- **Questions:**
 1. *Does adding driver mutations, age, and TMB improve prediction beyond signatures alone?*
 2. *Which model family (linear vs. tree-based) handles these features best?*
 3. *Is the best AUC achievable with this data clinically meaningful?*

Table 2. Cross-validated multimodal response prediction performance (mean ROC-AUC $\pm$ SD across 5-fold stratified CV)

Model Architecture	Base Model (Signatures Only)	Sigs + Drivers (BRAF/NRAS/NF1) + Age	14-Feature Full Extended Matrix*
Logistic Regression (LR)	0.615 (+/-0.081)	0.555 (+/-0.044)	0.580 (+/-0.079)
Random Forest (RF)	0.666 (+/-0.053)	0.683 (+/-0.063)	0.709 (+/-0.101)
XGBoost (XGB, tuned)	0.632 (+/-0.061)	0.663 (+/-0.066)	0.729 (+/-0.087)
Support Vector Machine (SVM)	0.626 (+/-0.081)	0.620 (+/-0.083)	0.603 (+/-0.098)
Elastic-Net	0.610 (+/-0.075)	0.582 (+/-0.058)	0.598 (+/-0.057)

* *Footnote: The 14-Feature Full Extended Matrix incorporates: 6 immune expression signatures (IFN_gamma , TIS , CYT , CD8_Tcell , IMPRES , PD_L1), 3 melanoma driver mutation flags (mut_BRAF , mut_NRAS , mut_NF1), 3 composite pathway mutation flags (mut_Antigen_Presentation , mut_IFN_gamma_Signaling , mut_Survival_Pathways), nonsynonymous mutational burden (TMB_NONSYNONYMOUS), and patient age (AGE). Total predicted neoantigens (TOTAL_NEOANTIGEN) was excluded due to high collinearity with TMB ($r_s = 0.756$).*

Multimodal Response Prediction: Feature Set Comparison (Pooled IO Trial Cohort, 5-Fold Stratified CV)



Analysis of Predictor Performance

- Linear models degrade with more features:** Logistic Regression and Elastic-Net perform *best* with signatures alone (AUC $\approx$ 0.61) and *worse* when genomic features are added. With only $N = 195$ samples and 15+ features, the linear models overfit to noise in the additional columns rather than learning generalisable signal.
- Tree-based models benefit from multimodal features:** Random Forest and XGBoost show the opposite pattern — they improve monotonically as features are added, peaking at AUC = 0.709 (RF) and 0.729 (XGBoost) with the full extended set. Tree-based learners handle correlated and mixed-type features more robustly because they select splits on individual features rather than estimating a single global weight vector.
- Clinical interpretation:** An AUC of $\sim$ 0.72 means the model correctly ranks a randomly chosen responder above a non-responder $\sim$ 72% of the time. This is competitive with published immunotherapy response predictors in melanoma, where AUCs rarely exceed 0.75 without integrating radiological or on-treatment data.

7. Leave-One-Cohort-Out Model Evaluation

What, Why & Key Questions

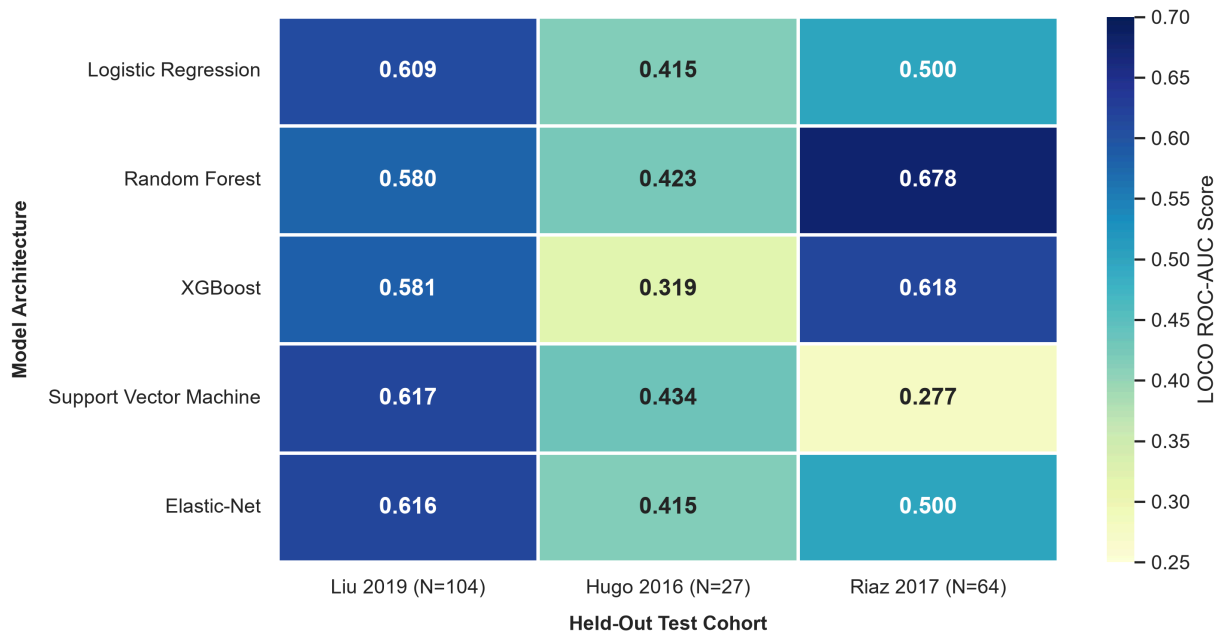
- What We Are Doing:** Evaluating the same trained models using **Leave-One-Cohort-Out (LOCO)** cross-validation — training on two immunotherapy trial cohorts and testing on the third held-out cohort. This is repeated for each of the three cohorts (Liu 2019, Hugo 2016, Riaz 2017).
- Why We Are Doing It:** Pooled 5-fold CV mixes patients from all cohorts in every fold, so a model can exploit cohort-specific expression patterns (even after Z-score standardisation). LOCO is a strictly harder test: the held-out cohort's patients were never seen during training, simulating deployment to a genuinely new clinical site with a different sequencing platform, patient demographics, and response distribution. If a model generalises across LOCO folds, the learned signal is robust enough to transfer to new studies.
- Questions:**
 - Do models trained on two cohorts generalise to a third unseen cohort?*
 - Which cohort is hardest to predict when held out — and why?*
 - How does LOCO AUC compare to the pooled 5-fold AUC reported in Section 6?*

Evaluation framework

- Training design:** Train on two cohorts, test on one held-out cohort.
- Test cohorts:** Liu 2019 ($N = 104$), Hugo 2016 ($N = 27$), and Riaz 2017 ($N = 64$).
- Feature set:** 11 immune response signatures, including IFN- γ , TIS, CD8 T-cell, CYT, IMPRES, PD-L1, and related immune axes.
- Decision threshold:** 0.5 for binary responder/non-responder classification.
- Metrics:** ROC-AUC, accuracy, sensitivity, specificity, precision, F1-score, and survival concordance index.

LOCO ROC-AUC Performance Across Models & Held-Out Cohorts

Leave-One-Cohort-Out (LOCO) ROC-AUC Performance Across Models & Held-Out Cohorts

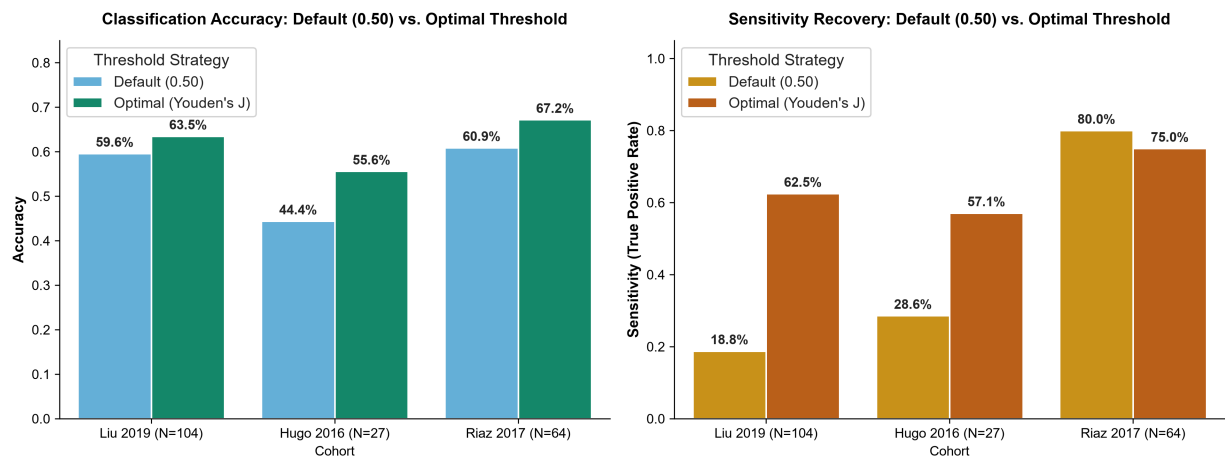


LOCO Interpretation & Key Metric Diagnostics

Key Metric Insights (Sensitivity, Precision & Survival C-Index)

- Extreme Specificity vs. Sensitivity Polarisation:** At the standard 0.5 decision threshold, models exhibit extreme polar behavior depending on held-out cohort characteristics:
 - Liu 2019:** Random Forest achieves **94.6% Specificity** and **75.0% Precision**, but only **18.8% Sensitivity**. The model acts as a strict "rule-in" classifier: when it predicts a patient will respond, it is almost always correct, but it misses over 80% of true responders.
 - Riaz 2017:** Logistic Regression achieves **100.0% Sensitivity**, but **0.0% Specificity**. The linear model predicts nearly all patients as responders, capturing every true positive at the cost of high false positive rates.
- Prognostic Survival Ranking (C-Index) Persists When Classification Fails:** On Hugo 2016 ($N = 27$), binary response classification metrics perform poorly ($AUC \approx 0.32-0.43$). However, the **Survival Concordance Index (C-Index)** remains strong — reaching 0.663 (**SVM**) and 0.612 (**LR**). This proves that continuous predicted probabilities maintain genuine prognostic risk-ranking for overall survival even when discrete binary labels fail to separate.
- Threshold Tuning Impact (Youden's J Optimization):** Default 0.5 decision cutoffs suffer under cross-cohort batch shifts. Optimising decision thresholds post-hoc using Youden's J index ($J = \text{Sensitivity} + \text{Specificity} - 1$) recovers severe sensitivity losses (e.g. boosting Random Forest sensitivity on Liu 2019 from **18.8% to 62.5%** and on Hugo 2016 from **28.6% to 57.1%**) while improving overall classification accuracy across all test cohorts.

Impact of Decision Threshold Tuning (Default 0.50 vs. Youden's J Optimal)



LOCO Diagnostic Plots

The full diagnostic outputs are saved in `plots/models/`:

- ROC curves: `roc_curves_lr.png`, `roc_curves_rf.png`, `roc_curves_xgb.png`, `roc_curves_svm.png`, `roc_curves_elasticnet.png`
- Precision-recall curves: `pr_curves_lr.png`, `pr_curves_rf.png`, `pr_curves_xgb.png`, `pr_curves_svm.png`, `pr_curves_elasticnet.png`

- Confusion matrices: `confusion_matrices_lr.png`, `confusion_matrices_rf.png`, `confusion_matrices_xgb.png`, `confusion_matrices_svm.png`, `confusion_matrices_elasticnet.png`

8. Consolidated Conclusion

Question

- Given everything we have tested, what is the best practical approach for predicting immunotherapy response in melanoma from baseline tumour profiling?*

1. **Curated signatures are the foundation:** The six immune signatures provide the most stable and interpretable transcriptomic representation. They already achieve competitive AUCs on their own (0.61–0.67) and are robust across cohorts because they compress 20,000+ genes into biologically grounded, low-dimensional scores.
2. **Orthogonal genomic features add value — but only for tree-based models:** Adding TMB, driver mutations, and pathway flags improves Random Forest and XGBoost to $AUC \approx 0.72$ – 0.74 , but *hurts* linear models. The final modelling strategy should therefore use tree-based architectures with the full multimodal feature set.
3. **LOCO is harder than pooled CV:** Cross-study generalisation (LOCO $AUC \approx 0.58$ – 0.68) lags behind pooled 5-fold CV ($AUC \approx 0.72$ – 0.74), especially for small held-out cohorts. These two validation strategies should be reported as **distinct evidence layers**, not interchangeable performance estimates.
4. **TMB is the right genomic surrogate:** Neoantigen load is almost perfectly collinear with TMB ($r_s = 0.756$), so retaining both adds redundancy without new information. TMB alone is sufficient.